# Arshia Akhtarkavan

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## Education

2023 – Expected 2027 Bachelor of Science, **University of Arizona**, Tucson, AZ (GPA: 4.0/4.0)
*W.A. Franke Honors College*
Majors: Physics, Mathematics (Comprehensive Emphasis), & Astronomy

2022 – 2023 Diploma, **Saba International High School**, Frankfurt, Germany

2020 – 2022 High School, **The Atomic Energy Institute**, Tehran, Iran

## Research Experience

**Undergraduate Researcher in Astrophysics** Nov 2023 – Present

*Advisor: Prof. Eiichi Egami (EGAMI Group) | Steward Observatory, University of Arizona*

**Galaxy Kinematics and Morphology**

**Research:** Investigate the kinematics and morphology of distant emission-line galaxies using JWST/NIRCam slitless spectroscopy. Developed and applied forward models using orthogonal grisms to infer galaxy rotation curves and morphology.

**Outcomes:** First-author manuscript in preparation; presented findings at AAS 247 and other conferences.

**High-Redshift Transients and Variability**

**Research:** Searched multi-epoch JWST/NIRCam imaging for transient and variable sources, identifying and characterizing candidates in distant galaxies.

**Outcomes:** Coauthored a peer-reviewed publication on a $z > 5$ transient/variable source (*ApJ*, 2025) and a TNS AstroNote reporting 39 transients/variables (2024).

**Additional contributions:** Processed JWST/NIRCam imaging and slitless spectroscopy; developed deep-learning methods for resolving ambiguous spectroscopic redshift solutions; contributed to the SAPPHIRES Early Data Release (*ApJS*, accepted).

**Dorrit Hoffleit Research Scholar** Jun – Aug 2026

*Advisors: Prof. Marla Geha (Geha Group) & Prof. Earl Bellinger (YAML Group) | Yale University*

**RR Lyrae Stars in the Draco Dwarf Galaxy**

**Research:** Investigated the properties and dynamics of RR Lyrae stars in Draco using spectroscopy and stellar-evolution modeling. Developed Bayesian methods for radial-velocity inference and distance estimation, aiming to constrain the galaxy's internal dynamics.

**Outcomes:** Presented research at two Yale workshops and symposia; analysis ongoing toward a research manuscript.

**Researcher in Mathematical Physics** Oct 2024 – Jan 2026

*Mathematics Department, University of Arizona (Advisor: Prof. Sergey Cherkis)*

**The Islands Project – Nahm Equations**

**Research:** Investigated the mathematical structure of Nahm equations, focusing on the construction of higher-order solutions and connections to gauge theory.

**Outcomes:** Produced an independent research report, presented the project at a university research forum, and received a competitive undergraduate research grant. Also rebuilt and expanded the Islands Project website to support student engagement in mathematical physics research.

## Publications

1. Forward-Modeling Galaxy Kinematics and Morphology with Orthogonal JWST/NIRCam Grisms in the SAPPHIRES Early Data Release
**Akhtarkavan A.**, Liu Y., Egami E., Sun F., Zhu Y., et al. (manuscript in preparation).

2. Hallmark: Git-style version control for scientific data catalogues
**Akhtarkavan A.**, Chan C.-k., Muthukumarasamy R. A., Wu T., et al. (manuscript in preparation).

3. The First Photometric Evidence of a Transient/Variable Source at $z > 5$ with JWST
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2025, *ApJ*, 990, 31; arXiv:2504.17007).

4. Slitless Areal Pure-Parallel HIgh-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field
Sun F., Fudamoto Y., Lin X., Helton J. M., Hsiao T. Y-Y., Egami E., **Akhtarkavan A.**, et al. (*ApJS*, accepted; arXiv:2503.15587).

5. Discovery of 39 Transients/Variables ($z_{\rm host} = 0.437$–$5.271$) in GOODS-N using JADES and CONGRESS JWST/NIRCam Images
DeCoursey C., Egami E., Sun F., **Akhtarkavan A.**, et al. (2024; Transient Name Server AstroNote 264).

## Conferences and Workshops

1. **"Forward-Modeling Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Flash Talk and Poster Presentation, *Surveying the Universe in 4D: Beating Cosmic Variance with Wide-Field Slitless Spectroscopy from HST, JWST, Euclid, Roman, and Beyond* Workshop
Space Telescope Science Institute (STScI), Baltimore, MD, August 2026

2. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Yale/CeNAM Nuclear Asteroseismology Workshop*
Yale University, New Haven, CT, July 2026

3. **"Modeling RR Lyrae Spectra in the Draco Dwarf Galaxy"**
Contributed Talk, *Summer Research Symposium*
Yale University, New Haven, CT, July 2026

4. **"Islands Project and the Nahm Equations"**
Talk, *Third Annual Cross-Campus Undergraduate Research Forum and Questionarium*
University of Arizona, Tucson, AZ, April 2026

5. **"Forward-Modeling Emission-Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
Contributed Talk, *Steward Observatory Internal Symposium*
Steward Observatory, University of Arizona, Tucson, AZ, March 2026

6. *Arizona Winter School on Multi-Probe Cosmology in the Roman ST Era*
Arizona Cosmology Lab, University of Arizona, Tucson, AZ, January 2026

7. **"Forward-Modeling Emission Line Galaxy Kinematics and Morphology with JWST NIRCam Grisms"**
iPoster presentation, *247th Meeting of the American Astronomical Society (AAS 247)*
Phoenix, AZ, January 2026

## Awards and Scholarships

Dorrit Hoffleit Research Scholarship, Yale University (2026)
Evelyn O. Bychinsky Promising Astronomer Award (2026)
William F. Lucas Astronomy Scholarship, Steward Observatory (2026)
College of Science Galileo Circle Scholar (Mathematics), University of Arizona (2026)
College of Science Galileo Circle Scholar (Physics), University of Arizona (2026)
Weaver Award for Undergraduate Research (2026)
Kenneth S. Krane Physics Scholarship (2025)

Evelyn O. Bychinsky Promising Astronomer Award (2025)
RII-Sponsored Campuswide Undergraduate Student-Initiated Original Research (2025)
Glenn C. Purviance Scholarship in Physics (2023)
Academic Year Highest Academic Distinction (2023 – Present)
Dean's List with Distinction - Full Time (2023 – Present)
Global Wildcat Tuition Award, University of Arizona (2023 – Present)
Silver Medal, National Astronomy and Astrophysics Olympiad (2022)

## Independent Studies and Graduate Coursework

**Quantum Field Theory** Fall 2026, in progress

*Physics Department, University of Arizona* (Advisor: Prof. Meinel)

– Graduate-level independent study

– Completing weekly problem sets and a final-term project

**Nahm Equations and Mathematical Physics** Fall 2025

*Mathematics Department, University of Arizona* (Advisor: Prof. Cherkis)

– Studied Nahm equations and related topics in gauge theory and representation theory

– Wrote a final report on the construction of higher-order Nahm solutions

**Graduate-Level Coursework**
ASTR 540: Structure and Dynamics of Galaxies (Prof. Peter Behroozi; in progress)
PHYS 515B: Electromagnetic Theory (Prof. Stefan Meinel; in progress)
PHYS 569: Introduction to General Relativity (Prof. Vasileios Paschalidis)

## Professional Experience

**Astronomical Software Engineering Intern** Aug 2026 – Present

*TIMESTEP, University of Arizona* (Event Horizon Telescope Group; Supervisor: Dr. CK Chan)

– Participating in an academic-year internship focused on developing `Hallmark`, a lightweight Python package for version-controlling and distributing scientific data products across Event Horizon Telescope analysis pipelines.

**Mechanical Engineering Workshop Intern** Jan – Mar 2023

*FH Aachen University, Essen, Germany*

– Gained hands-on experience operating machining and workshop equipment and interpreting and producing technical schematics.

## Teaching and Outreach

**Mathematics Grader** Aug 2025 – May 2026

*Mathematics Department, University of Arizona, Tucson, AZ*

– Graded assignments and examinations, prepared detailed solutions, and provided written feedback for upper-division courses in differential equations and linear algebra.

**Undergraduate Mentor** Oct 2026 – Present

*University of Arizona, Tucson, AZ*

– Participate in an undergraduate mentoring program, providing peer guidance and support to one student.

**Astronomy Olympiad Teaching Assistant** Oct 2022 – Aug 2023

*The Atomic Energy Institute, Tehran, Iran*

– Taught cosmology and prepared lecture notes for Astronomy Olympiad students, based on textbooks by Ryden and Carroll.
– Selected problems and led problem-solving sessions in celestial mechanics.

**Educational Materials Translator** 2025

*Arizona Astrobiology Center, University of Arizona, Tucson, AZ*

– Translated Astrobiology lecture materials from English into Farsi for Persian-speaking elementary school students

## Skills

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|---|---|
| Programming Languages | Python, C/C++, SQL |
| Scientific & Data Analysis Libraries (Python) | NumPy, Matplotlib, PyTorch, Astropy, SciPy, Specutils, Photutils, Pandas, JAX |
| Astronomy & Astrophysics Software | DS9, JWST Pipeline, Source detection and Photometry, Webbpsf, NIRCam imaging and grism spectroscopy |
| Software & Research Systems | Linux/Unix, Git/GitHub, HPC environments, remote computing (ssh), GPU-accelerated computing (CUDA), Jupyter, VS Code, Emacs, LaTeX, Microsoft Office |
| Concepts | Data Processing Pipelines, Scientific Computing, Machine Learning, Numerical Methods |

## Languages

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|---|---|
| English | Fluent |
| Farsi (Persian) | Native |
| German | B1 level |
| Mandarin Chinese | Currently learning |